Duke University **November 10, 2020**
Math 105L: Laboratory Calculus and Functions I

Big Assignment 3

- This assignment covers work from Day 10-3 to Day 13-3.
- You have ten days to complete the assignment. It is due Nov 20 at 8am.
- You may use the worksheets, labs, and your notes, as well as any other resources.
- However, you may not work with anyone else on this assignment – it must be entirely your own work.
- Throughout the assignment, show your work so that your reasoning is clear. Otherwise no credit will be given. Circle your answers. For example, copying graphs from graphing software is insufficient 'work'.
- Do not spend too much time on any one problem. Read them all through first and attack them in an order that allows you to make the most progress.
- You must hand in a scanned version of this assignment on Gradescope and no other pages. If you need to do rough work, do it elsewhere before writing your answers.
- You must sign the statement at the bottom of this page.

Honor Code:

I have adhered to the Duke Community Standard and all the rules above in completing this assignment.

Signature: _____

Question 1 (15 points) Consider the function $f(x) = 4\ln(x) + (x-3)^2$, with domain $0 < x \leq 5$.

(a) Find and classify all critical points of this function.

(b) Does this function have a global min and/or a global max? For each, either find it or explain why it does not exist.

(c) Find all inflection points of $f(x)$. Be sure to check that each is in fact an IP!

Question 2 (15 points) The following are false. Give a counterexample in each case. In some cases, a good picture will suffice. In others, an explicit formula may be needed.

(a) If $f''(a) = 0$, then $f(x)$ has an inflection point at $x = a$.

(b) If $f''(a) > 0$, then $f(x)$ has a local minimum at $x = a$.

(c) A continuous function with domain all real numbers cannot have both a global minimum and a global maximum.

(d) If a differentiable function $f(x)$ has $f(0) = f(1) = 0$, $f'(x)$ cannot have exactly two roots between $x = 0$ and $x = 1$.

(e) If a continuous function $f(x)$ has $f(0) = f(1) = 0$, then $f'(x)$ must have at least one root between $x = 0$ and $x = 1$

Question 3 (15 points) This question is about linear approximations of $f(x) = \sqrt{x}$.

(a) By computing derivatives, show that $f(x)$ is increasing and concave down.

(b) Do you expect linear approximations of values of $\sqrt{2.5}$ using nearby points to be overestimates, underestimates, or neither?

(c) Show that the tangent lines to $f(x) = \sqrt{x}$ at $x = 1$ and $x = 4$ are $\frac{1}{2}x + \frac{1}{2}$ and $\frac{1}{4}x + 1$ respectively.

(d) Compute the linear approximations of $\sqrt{2.5}$ using each of your tangent lines from the previous part of this question.

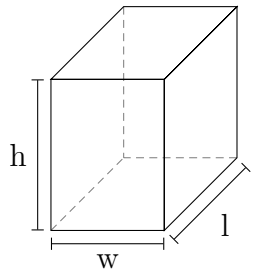
(e) Note that $x = 2.5$ is exactly half way between $x = 1$ and $x = 4$. Compare your answers from the previous part of this question to the real value of $\sqrt{2.5}$. Which approximation is better?

(f) Can you explain your answer to the previous part of this question? Why aren't the two approximations equidistant from $\sqrt{2.5}$? (Hint: Consider $f''(x)$, and perhaps draw a picture.)

Question 4 (12 points) The US Postal Service specifies that a box to be sent through the mail must satisfy

$$\text{girth} + \text{length} \leq 108\text{in},$$

where girth is the sum of the four sides other than length. That is, girth is $2h + 2w$:



If the front is square (i.e. $h = w$), find the dimensions that maximize the box's volume.

Question 5 (6 points) Compute the following limits:

(a) $\lim_{x \to 0} \dfrac{x}{\ln(x+e)}$

(b) $\lim_{x \to 0^+} x \ln x$

Question 6 (14 points) Order the following functions in order of dominance:

$$x^2, x^2 + 2x, e^x, e^{x^2}, e^{x+2}, \ln x, 5^x.$$

You may appeal to 'family dominance' if you wish (and thus avoid using L'Hôpital's rule). Within families, however, you must show limit computations. Certain pairs of functions may be equivalent under dominance (i.e. neither dominates the other.)

Question 7 (13 points) Consider a square-based pyramid. The volume of a square-based pyramid is

$$V = \frac{1}{3}b^2h,$$

where b is the length of one side of the pyramid's base, and h is the pyramid's height, measured from the center of its base to its pinnacle.

(a) Suppose that the length of the sides of the base are increasing at 3 cm/min when they are 30 cm long, and that the height is decreasing at a rate of 2 cm/min then. Suppose that at this point in time, the pyramid's volume is $3,000$ cm^3. Find the rate at which the volume is changing.

(b) Suppose instead that the volume is decreasing at 2 cm^3 /min when the height is 9 cm and the base length is 6 cm. Express $\frac{dh}{dt}$ in terms of $\frac{db}{dt}$.

Bonus Question (1 point) Where do you see yourself in five years from now? What will you be doing?

Bonus Question (1 point) Tell me a joke, or draw a funny picture.